THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 10: **Circles** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1	Exactly one tangent at a point
2	**(a)** 15 cm	$\sqrt{289-64}=15$
3	**(b)** secant	Definition
4	**(c)** 90°	Theorem 10.1
5	**(b)** Incorrect — no tangent can be drawn from an interior point	Every line through an interior point is a secant
6	**(b)** 110°	$\angle POQ = 180^\circ - 70^\circ$
7	**(a)** Student A is correct	$OP \perp PQ \Rightarrow \sqrt{169-25}=12$
8	**(a)** 10 cm	$PA = PB$ and $\angle P = 60^\circ \Rightarrow$ equilateral
9	**(b)** parallel	Both are perpendicular to the same diameter
10	**(a)** 3 cm	$AB + CD = BC + AD \Rightarrow 10 = 7 + AD$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Tangents at A and B are $\perp$ to diameter AB (Th. 10.1). Two lines perpendicular to the same line make equal alternate angles (90°) $\Rightarrow$ parallel.	1+1
12	$\angle OAT = 90^\circ \Rightarrow \angle OAB = 30^\circ = \angle OBA$ ($OA = OB$) $\Rightarrow \angle AOB = 120^\circ$	1+1
13	In quadrilateral OPTQ, $\angle OPT = \angle OQT = 90^\circ$; angle sum $360^\circ \Rightarrow \angle POQ + \angle PTQ = 180^\circ$	1+1
14	False. $\triangle OPT \cong \triangle OQT$ (RHS: $OP = OQ$, OT common, right angles) $\Rightarrow TP = TQ$ always.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Given, To prove, figure. In $\triangle OPT, \triangle OQT$: $OP = OQ$, OT common, $\angle OPT = \angle OQT = 90^\circ \Rightarrow$ congruent (RHS) $\Rightarrow TP = TQ$	1+2
16	Radius to point of contact is $\perp$ to chord and bisects it: half chord $= \sqrt{169-25} = 12 \Rightarrow$ chord $= 24$ cm	1+1+1
17	Let the sides touch at P, Q, R, S. Tangents from a point are equal: $AP = AS,\ BP = BQ,\ CR = CQ,\ DR = DS$. Adding: $AB + CD = AD + BC$	1+1+1
	OR: $AB + CD = AD + BC$ (tangent property); $AB = CD,\ AD = BC \Rightarrow 2AB = 2AD \Rightarrow$ all sides equal $\Rightarrow$ rhombus	1+1+1
18	$TP = TQ \Rightarrow \angle TPQ = \angle TQP = 90^\circ - \frac{1}{2}\angle PTQ$. $\angle OPQ = 90^\circ - \angle TPQ = \frac{1}{2}\angle PTQ \Rightarrow \angle PTQ = 2\angle OPQ$	1+1+1
	OR: Let $AF = AE = x$; $BD = BF = 11 - x$, $CD = CE = 8 - x$. $BD + DC = 9 \Rightarrow 19 - 2x = 9 \Rightarrow x = 5 \Rightarrow AF = 5,\ BD = 6,\ CE = 3$ cm	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Take any point Q on tangent XY other than P; Q lies outside the circle $\Rightarrow OQ > OP$. So OP is the shortest distance from O to XY $\Rightarrow OP \perp XY$ (3). Tangent $= \sqrt{225-81} = 12$ cm (2)	3+2
	OR: Let $AF = AE = x$: $AB = x + 8,\ AC = x + 6,\ BC = 14,\ s = x + 14$. Area $= r \cdot s = 4(x+14) = \sqrt{(x+14)\cdot x \cdot 6 \cdot 8} \Rightarrow x + 14 = 3x \Rightarrow x = 7 \Rightarrow AB = 15$ cm, $AC = 13$ cm	2+2+1
20	(i) $\sqrt{625-49} = 24$ m (ii) tangents from an external point are equal (iii) 90° (iv) $7 + 24 + 24 + 7 = 62$ m	2+1+1+1
	OR: In OPTQ, $\angle OPT = \angle OQT = 90^\circ \Rightarrow \angle PTQ + \angle POQ = 360^\circ - 180^\circ = 180^\circ$ (3). $\angle PTQ = 55^\circ$ (2)	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 10: **Circles** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 2	At the two ends of the diameter perpendicular to that direction
2	**(a)** 8 cm	$\sqrt{100-36}=8$
3	**(b)** point of contact	Definition
4	**(a)** 40°	$180^\circ-140^\circ$
5	**(b)** Incorrect — a line through the centre meets the circle at two points	A line through the centre is a secant (diameter line)
6	**(a)** 25°	$\angle PAB=65^\circ$; $\angle OAB=90^\circ-65^\circ$
7	**(a)** Student A is correct	$AB+CD=BC+AD \Rightarrow AD=6$
8	**(a)** 3 cm	$\sqrt{25-16}$
9	**(b)** 8 cm	Equal to the diameter
10	**(a)** 3 cm	OAPB is a square

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\angle QOR=180^\circ-46^\circ=134^\circ$	1+1
12	Half chord $=\sqrt{100-36}=8$ ⇒ chord $=16$ cm	1+1
13	The radius at the point of contact is $\perp$ to the tangent; only one perpendicular can be drawn at a point on a line ⇒ the perpendicular coincides with the radius, so it passes through O.	1+1
14	$\angle OAP=\angle OBP=\angle APB=90^\circ$ and $OA=OB$ ⇒ OAPB is a square ⇒ $PA=5$ cm. ($5\sqrt{2}$ is OP.) Claim is false.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Given, To prove, figure. Any point Q ($\neq P$) on the tangent is outside the circle ⇒ $OQ>OP$ ⇒ OP is the shortest distance ⇒ $OP\perp$ tangent	1+2
16	OA bisects $\angle PAC$ and OB bisects $\angle QBC$ (equal-tangent congruences). $\angle PAB+\angle QBA=180^\circ$ (co-interior) ⇒ $\angle OAB+\angle OBA=90^\circ$ ⇒ $\angle AOB=90^\circ$	1+1+1
17	Joining O to the vertices and points of contact gives 8 angles in 4 equal pairs; $2(a+b+c+d)=360^\circ$ ⇒ $\angle AOB+\angle COD=a+b+c+d=180^\circ$	1+1+1
	OR: OP $\perp$ AB (radius $\perp$ tangent). The perpendicular from the centre to a chord bisects the chord ⇒ $AP=PB$	1+1+1
18	Tangents at P and Q meet at T. $TP=TQ$ ⇒ $\triangle TPQ$ is isosceles ⇒ $\angle TPQ=\angle TQP$	1+1+1
	OR: $PA=PB$, $\angle P=60^\circ$ ⇒ equilateral ⇒ $AB=12$ cm; $\angle PAB=60^\circ$ ⇒ $\angle OAB=90^\circ-60^\circ=30^\circ$	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	RHS congruence ⇒ equal tangents (3). $AQ=AR$, $BQ=BP$, $CR=CP$: $2AQ=AQ+AR=AB+BP+AC+CP=AB+BC+CA$ (2)	3+2
	OR: $AD=AF=x$, $BD=BE=12-x$, $CE=CF=10-x$; $BE+EC=8 \Rightarrow x=7$ ⇒ $AD=7$, $BE=5$, $CF=3$ cm (4). Sum $=15=\frac{30}{2}$ ✓ (1)	4+1
20	(i) $25=25$ ✓ (ii) $BP=BQ=6$, $CQ=CR=9$, $DR=DS=6$ (and $AS=4$ ✓) (iii) Equal tangents from each vertex; add them	1+2+2
	OR: (i) $TP=TQ$ and $OP=OQ$ ⇒ both O and T lie on the perpendicular bisector of PQ (3) (ii) $TP=12$; $PR=\frac{OP\cdot TP}{OT}=\frac{60}{13}$ ⇒ $PQ=\frac{120}{13}$ cm (2)	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.